

Answer Key — Biology Grade 11

Sub-Strand 2.2: Growth and Development in Plants • Lesson 1: The Farmer's Puzzle: Why Do Maize and Beans Behave Differently After the Same Rain?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
D	C	B	A	C	A	B

USE AFTER: PREDICT PHASE — Predict: the farmer's story about Mama Wanjiru in Molo

Q1. Mama Wanjiru in Molo plants maize and beans on the same morning after the first heavy rains. The seeds all look healthy. After one week, which description matches what the story says she found?

- A. All the seeds came up on the same day
- B. Every healthy-looking seed had sprouted
- C. Only the beans came up and every maize seed failed
- D. Some seeds had sprouted, some were late, and a few had not come up at all**

Correct answer: D. The story says that after one week some seeds had sprouted, some were late and a few had not come up, even though they looked healthy. This is the puzzle the class will explore.

USE AFTER: OBSERVE PHASE — Observe: 'I notice / I wonder' table for the bean and maize time-lapse videos

Q2. While watching a time-lapse of a bean seed growing in soil, four learners wrote notes. Which note is a specific observation, something you can actually see, and not a guess or an explanation?

- A. The bean seed needs warmth to wake up.
- B. Maize and beans grow in exactly the same way.
- C. The bean stem was bent like a hook as it came above the soil.**
- D. The plant grew.

Correct answer: C. A specific observation describes something seen, such as the hooked stem. 'The plant grew' is too general, and the other two are explanations or claims, not observations.

USE AFTER: OBSERVE PHASE — Observe: second viewing of the time-lapse videos, recording 'I notice' and 'I wonder'

Q3. In the time-lapse videos, learners were asked to watch which part of the seedling came out first. Which statement is an 'I wonder' statement rather than an 'I notice' statement?

- A. I notice that the maize leaf was rolled up.
- B. I wonder why some seeds that looked healthy did not sprout.**
- C. I notice that some seeds did not sprout.
- D. I notice that the root grew down into the soil.

Correct answer: B. An 'I wonder' statement raises a question about something not yet understood. The other statements describe what can be seen.

USE AFTER: EXPLAIN PHASE — Explain: drawing and labelling the initial model of a maize seed and a bean seed

Q4. Your teacher says your first drawn model of a maize seed and a bean seed will not be marked right or wrong, and you will keep it in a folder. What is the main reason for this?

- A. It shows your starting ideas, so you can compare them with your improved ideas at the end of the unit**
- B. It will be used to give each learner a final grade for the term
- C. Only learners who draw the correct model are allowed to continue
- D. The teacher already knows that every first model is wrong and will discard them

Correct answer: A. The unmarked Day 1 model is a baseline of honest thinking. Learners return to it at the end of the unit to see how their ideas changed with evidence.

USE AFTER: EXPLAIN PHASE — Explain: sentence starters 'I think maize and bean plants grow differently because...' and 'I wonder why...'

Q5. On Day 1 of the unit, a learner asks: 'Why did some of Mama Wanjiru's healthy-looking seeds not germinate?' Which is the best response at this stage of learning?

- A. Only the biggest seeds germinate, so the small ones failed.
- B. Healthy-looking seeds always germinate, so the farmer must have counted wrongly.
- C. It has not been explained yet, so it goes on the Driving Question Board to investigate with evidence.**
- D. Seeds need only water, so the rain should have made every seed germinate.

Correct answer: C. This is the anchor lesson, so the phenomenon is left unexplained. The other choices are the misconceptions 'healthy seeds must germinate', 'seeds need only water' and 'big seeds grow first'.

USE AFTER: DQB PHASE — DQB: groups write and improve questions on sticky notes or cards

Q6. A group wants to post a question on the Driving Question Board about maize and beans. Which question is the best one to investigate?

- A. How does maize grow differently from beans?**
- B. Is maize good?
- C. Is Mama Wanjiru a good farmer?
- D. Did it rain in Molo?

Correct answer: A. A good DQB question starts with How, Why or What and can be investigated. The other questions are yes/no or cannot be answered by studying how the plants grow.

USE AFTER: END OF LESSON — End of lesson: choosing the first new question from the DQB and the exit ticket link to the driving question

Q7. Many groups asked whether growth and development are the same thing. Which question did the class agree to investigate first in the next lesson?

- A. Why is the rainfall heavy in Nakuru County?
- B. What is growth and development?**
- C. Which fertiliser is best for maize?
- D. How do plants make their own food?

Correct answer: B. The question that appeared most often when the class sorted the DQB was 'What is growth and development?', so it leads into Lesson 2.